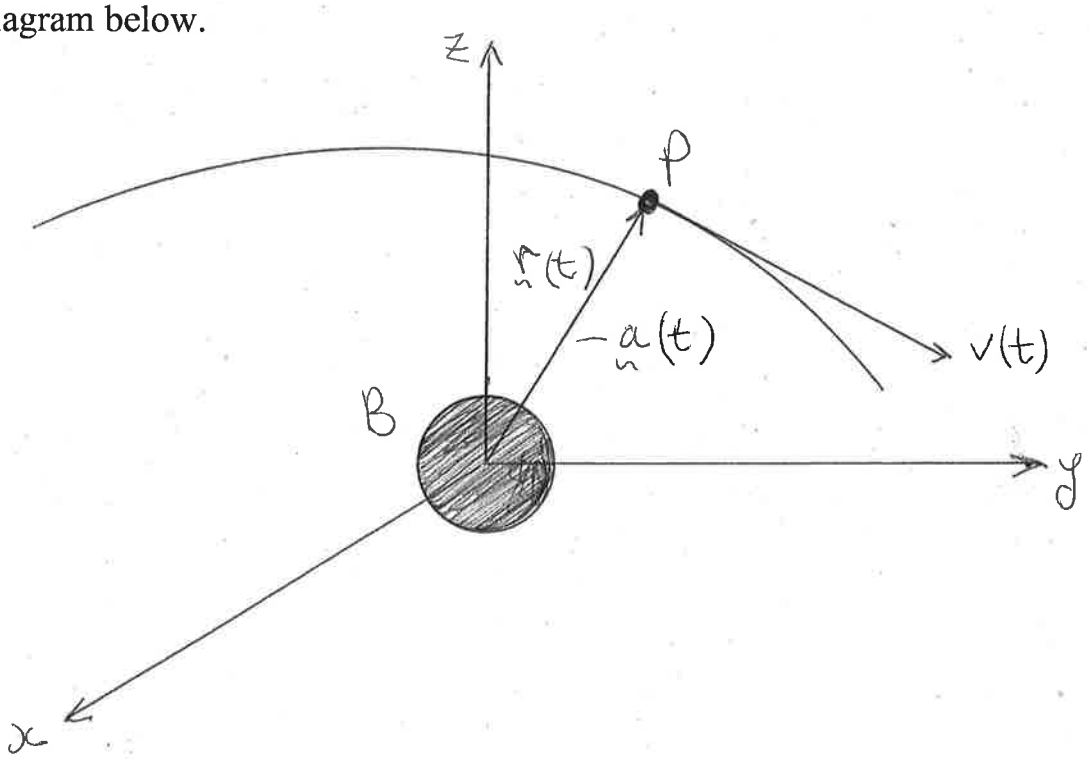
POAMS: LECTURE 16
POAMS AND CIRCULAR ORBITAL MOTION

In this lecture, I shall continue to apply the fundamental postulates of POAMS to the two-body problem. I will go on to study special cases of the POAMS equation of motion and to provide two specific numerical examples. This lecture is based on some of Chapter 6 of 'Light-speed....'.

1. The POAMS approach to the two-body problem [*Recap*]

In this lecture, we continue to apply POAMS (*the Pope-Osborne Angular Momentum Synthesis*) to the study of the *two-body problem*. Recall that this problem consists of a particle, P, of mass m orbiting a body, B, of mass M, with m very much less than M. As before, we let the centre-of-mass of B lie at the origin, O, of a set of Cartesian coordinates and let the position vector of P, in relation to O, at any particular time t be $\mathbf{r}(t)$, as shown in the diagram below.



The POAMS approach to this two-body problem uses a combination of conservation of angular momentum together with empirical evidence to derive the equation of motion obtained by Newtonian theory, without any need for an *in vacuo* 'gravitational force'.

Here, the velocity, $\mathbf{v}(t) = \mathrm{d}\mathbf{r}/\mathrm{d}t$, of P is tangent to its orbit. Since the *angular momentum*, $\mathbf{L}$, of the particle P relative to the chosen origin, O, is a constant, it follows that that the orbit of P *lies in a plane* and that $L = \|\mathbf{L}\|$ is a constant. It also follows that the acceleration of P, $\mathbf{a}(t) = \mathrm{d}\mathbf{v}/\mathrm{d}t$, is parallel to $\mathbf{r}(t)$ and is directed towards O.

Recall that the *angular speed* of P is $\omega(t) = \mathrm{d}\theta/\mathrm{d}t$, where θ is the angle between the radial vector $\mathbf{r}$ and some fixed radial axis. Exactly as in Section 3 of Lecture 13, the (orbital) speed, $v(t) = \|\mathbf{v}(t)\|$, of P relative to O is then given by

$$v^2 = \left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + r^2\omega^2 \quad (1)$$

where $r = \|\mathbf{r}(t)\|$ [*Lecture 15, equation (2)*] and the magnitude of P's angular momentum is

$$L = mr^2\omega \quad (2)$$

[*Lecture 15, equation (3)*]. The acceleration of P is given by

$$\mathbf{a}(t) = \left(\frac{\mathrm{d}^2r}{\mathrm{d}t^2} - r\omega^2\right)\mathbf{n} \quad (3)$$

where $\mathbf{n}$ is a vector of unit length in the direction of $\mathbf{r}$ [*Lecture 15, equation (4)*].

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2. <u>The POAMS equation of motion [*Recap*]</u>

Recall that, based on conservation of momentum and the empirical evidence alone, the POAMS second-order equation of motion for P orbit about B for which stable closed orbits are possible is,

$$\frac{\mathrm{d}^2r}{\mathrm{d}t^2} - \frac{L^2}{m^2r^3} = -\frac{\alpha}{r^2} \quad (4)$$

where α is a constant [*Lecture 15, equation (11)*]. This is a consequence of *Bertrand's Theorem.* As in Newtonian theory, the left-hand side of this equation gives the magnitude of *P*'s acceleration (see equations (2) and (3)). This includes the special cases of 'radial free-fall' when $L = 0$, so that θ is a constant, and circular motion, for which *r* is a constant. This equation is to be used in conjunction with

$$\omega = \frac{d\theta}{dt} = \frac{L}{mr^2} \qquad (5)$$

(see equation (2)) [*End of recap*].

Since we wish to compare equation (4) with the corresponding Newtonian equation of motion [*equation (9) of Lecture 13*], we shall re-label the constant α as *GM*, where *M* is the mass of *B* and *G* is simply a new constant, with no 'gravitational' connotations.

Recall that, according to the POAMS thesis, the true (*i.e.* measurable) force on the particle *P* is not *concealed* in its natural orbit but is *revealed* in the difference between the natural orbit and an 'unnatural' one, that is, some state of motion in which the particle is held by a measurable force, of magnitude *F*. In general, of course, *P* may be artificially made to follow any constrained orbit about *B* and so may not have an equation of the form of equation (4). But let us consider the specific case of a constrained orbit of *P* with an equation of motion of the form (4). In this case, we may let $\alpha = G'M$, where G' is a constant with the same dimensions as the Newtonian 'gravitational' constant, *G*, but only equal to *G* in any natural (unconstrained) orbit of *P*. Then the measurable force exerted on *P* in this constrained orbit has magnitude, *F*, given by

$$F = |G - G'|\frac{mM}{r^2} \qquad (6)$$

Otherwise, in a natural, 'force-free' orbit, $G' = G$ and $F = 0$.

For a particular constrained orbit of *P* of the type just described, the parameters $r(t)$ and $\omega(t)$ are known observational measures. Since from equations (2) and (4), we have, in this case,

$$\frac{d^2r}{dt^2} - r\omega^2 = -\frac{G'M}{r^2}$$

where the mass, M, of the body B is known, the constant G' for this orbit can be calculated. The magnitude, F, of the measurable force exerted on P in this constrained orbit is then given by equation (6).

If the particle P orbits the body B in a constrained bounded planar orbit for which the magnitude of P's angular momentum, L, is constant, then since $r(t)$ and $\omega(t)$ are known, L may be calculated from equation (2). It is then possible to derive any natural closed orbit of P, with the same magnitude of angular momentum L, using equation (4) with $\alpha = GM$ and equation (5). Notice that, in general, given L, any natural orbit of P is not unique, since it depends on a given set of initial conditions.

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3. Circular orbits

We now restrict our discussion to the special case of *circular* orbits, both natural and constrained, in order to illustrated a number of ideas in the simplest possible way. Such orbits occur when r is a constant in equation (4) and a particular value of r gives a particular circle of radius r. In this special case, the velocity, $\mathbf{v}(t)$, of P is *orthogonal to the position vector*, $\mathbf{r}(t)$ of P:

$$r^2 = \|\mathbf{r}\|^2 = \mathbf{r}.\mathbf{r} \Rightarrow 2r\frac{dr}{dt} = 0 = 2\frac{d\mathbf{r}}{dt}.\mathbf{r} \Rightarrow \mathbf{v}.\mathbf{r} = 0$$

It then follows from the definition of $\mathbf{L}$ that

$$L = \|\mathbf{L}\| = \|\mathbf{r} \times m\mathbf{v}\| = mvr \qquad (7)$$

This can also be seen from equation (1), which gives $v = r\omega$, together with equation (2). It then follows by equation (4) with $\alpha = GM$, any *natural circular* orbit satisfies

$$v^2 = \frac{GM}{r} \qquad (8)$$

so that there is a *unique* natural circular orbit for any given radius.

Within the context of the sorts of constrained closed orbits we have been considering, one of the most useful cases to consider is the case in which P is fixed to the surface of a rotating approximately spherical body B, such as Earth, and so can be considered to orbit the centre-of-mass, O, of B. In this case, P's constrained orbit is a circle, with centre O. It follows from equation (8) that in this special case, the constant, G', for this constrained orbit is

$$G' = \frac{rv^2}{M} \quad (9)$$

where v is the constant orbital speed of P in this orbit of radius r.

Given such a constrained orbit of P with constant angular momentum of magnitude L, there is a unique *circular* natural orbit for P, with the same magnitude of angular momentum, where from equations (7) and (8), the orbital speed, v_0, of P is given by

$$v_0 = \frac{GmM}{L} \quad (10)$$

[see endnote (*) for details] and the radius, r_0, of P's orbit is then given by

$$r_0 = \frac{L}{mv_0} \quad (11)$$

Notice that, from equations (10) and (11), or from equation (8), it follows that in any natural circular orbit,

$$v_0^2 = \frac{GM}{r_0} \quad (12)$$

independent of the mass m [see endnote (**) for details]. Increasing the radius r_0 will decrease the speed v_0 and decreasing r_0 will increase v_0.

In the examples that follow, the figures given are intended only to illustrate the key ideas. The constants quoted are the currently accepted values and differ slightly from the figures used in 'Light- Speed... '.

4. Example One

Consider a satellite P of mass 1 kg in a circular orbit about Earth, whose radius relative to the centre of Earth is $r_0 = 8 \times 10^6$ m. In order for P to follow a natural (force-free) orbit about Earth, its speed, v_0, must be given by equation (12). Using the currently accepted classical values of $G \cong 6.6743015 \times 10^{-11}$ m^3 s^{-2} kg^{-1} and $M \cong 5.9722 \times 10^{24}$ kg, this gives $v_0 \cong 7.058706 \times 10^3$ m s^{-1}. If the speed of P is reduced, then P will take up a natural orbit further from Earth if allowed to do so. On the other hand, in order for P to attempt to take up a natural orbit with radius $r_0^* = 6.371088 \times 10^6$ m, say, then its new speed will then be $v_0^* \cong 7.909760 \times 10^3$ m s^{-1}. Here, r_0^* just happens to be the currently accepted value of the overall mean radius of Earth, so that if the speed of P is increased to v_0^* the orbit of P, if allowed to do so, would decay and P would crash at a point on Earth's surface.

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5. Example Two

Consider a particle P of mass 1 kg, say, situated somewhere on Earth's equator. The speed of P in this constrained approximately circular orbit about Earth's centre is the rotational speed of Earth, namely $v \cong 465.111$ m s^{-1}. The radius of this orbit is Earth's mean equatorial radius, given by $r \cong 6.378137 \times 10^6$ m. For this constrained circular orbit, the constant G' is given by equation (9), where M is the mass of Earth. It is important to notice here that this is not exactly the usual figure quoted for M in Newtonian dynamics since in POAMS, according to equation (6),

$$\frac{M(G - G')}{r^2} = g$$

where g is the usual acceleration due to the classically conceived 'gravitational pull' of Earth. Hence, by equation (9), it follows that,

$$gr^2 = MG - rv^2 \Rightarrow M = r(rg + v^2)/G$$

where $g \cong 9.80665$ m s^{-1} and $G \cong 6.6743015 \times 10^{-11}$ m^3 kg^{-1} s^{-2}. Hence, $M \cong 5.997939 \times 10^{24}$ kg. It then follows by equation (9) that

$$G' = 2.3004099 \times 10^{-13}\ \mathrm{m^3\ kg^{-1}\ s^{-2}}$$

Note that this figure is independent of the mass of P. Since $G' \neq G$, it follows that the particle P is not in a natural orbit about the Earth's centre. It then follows by equation (6) that the magnitude of the real, measurable force on P at Earth's equator is then,

$$F = \frac{M(G - G')}{r^2} \cong 9.80665\ \mathrm{N} = 9.80665\ \mathrm{kg\ m\ s^{-2}}$$

[Here, N is the *Newton*, the SI unit of force, defined as the magnitude of the force required to accelerate a 1 kg mass up to 1 metre per second per second.] This is the magnitude of the force with which a 1 kg mass presses on Earth's surface which prevents it from taking up its natural, force-free orbit. It is the *real* force that produces a weight of 1 kg on a weighing-scale. Of course, this follows from Newton's second law with $g \cong 9.80665$ ms^{-1}.

Since the motion of P in its constrained orbit is approximately circular, the magnitude of its orbital angular momentum, L, is given by equation (7), so that

$$L = vr \cong 2.966542 \times 10^9\ \mathrm{kg\ m^2\ s^{-1}}$$

With this relatively small amount of angular momentum, if we imagine Earth's radius shrunk to the size of a super-dense billiard ball, the natural orbital speed, v_0, of that kilogram mass, assuming its natural orbit to be circular around Earth's mass, is given by equation (10), so that

$$v_0 = \frac{GM}{L} \cong 1.349452 \times 10^5\ \mathrm{m\ s^{-1}}$$

The radius, r_0, of this natural orbit is then given by equation (11) so that

$$r_0 = \frac{L}{v_0} \cong 2.198332 \times 10^4\ \mathrm{m}$$

This radius is approximately 1/290th of Earth's actual radius, so it follows that the mass would seek to orbit at 289/290^{ths} of the distance down from the Earth's surface to its centre, if allowed to do so. Note that, once again, the values of v_0 and r_0 do not depend on the mass of P.

In the next lecture, we shall continue to study constrained and natural circular orbits in POAMS and I will provide a further numerical example, before moving on to discuss the POAMS analogue of Newton's Second Law.

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(*) From eqn. (7), $L = mv_0 r_0$

Then from eqn. (8), $L = mv_0 \left(\frac{GM}{v_0^2} \right) = \frac{GmM}{v_0}$ (10)

(**) From eqn. (10), $v_0 = \frac{GmM}{L}$

Then from eqn. (11), $v_0 = \frac{GmM}{mv_0 r_0}$

$\Rightarrow v_0^2 = \frac{GM}{r_0}$ (12)

Eqn. (12) also follows directly from eqn. (8)
